Check for updates

Remote Sensing in Ecology and Conservation
Open Access
ZSL

RESEARCH ARTICLE

Programmed unmanned aerial vehicles show great potential for monitoring marine megafauna in specific areas of interest

Dinah Hartmann[1,2,3], Valdemar Palmqvist[1] & Johanna Stedt[1]

[1]Department of Biology, Lund University, Lund, Sweden
[2]Centre for Research into Ecological and Environmental Modelling, School of Mathematics and Statistics, University of St Andrews, St Andrews, UK
[3]Scottish Oceans Institute, University of St Andrews, St Andrews, UK

Keywords
conservation, density, digital aerial surveys, drone, habitat use, monitoring, *Phocoena phocoena*

Correspondence
Johanna Stedt, Department of Biology, Lund University, Ecology Building, Naturvetarvägen 6A, 223 62 Lund, Sweden. E-mail: johanna.stedt@biol.lu.se

Funding Information
This work was supported by the Royal Physiographic Society of Lund (grant no. 39589 to JS) and The Royal Swedish Academy of Sciences (grant no. BS2018-0080 to JS).

Editor: Dr. Temuulen Sankey
Associate Editor: Dr. Michael Espriella

Received: 30 June 2025; Revised: 25 October 2025; Accepted: 29 October 2025

doi: 10.1002/rse2.70043

Abstract

Targeted conservation measures are contingent on robust knowledge of spatio-temporal animal distribution in areas of interest. We explore unmanned aerial vehicle (UAV) transect monitoring as a novel method for standardized digital aerial surveys of marine megafauna by investigating the fine-resolution spatio-temporal distribution of harbour porpoises (*Phocoena phocoena*) in a Swedish nature reserve along with drivers of this distribution and potential biases. Biweekly UAV video data were collected along pre-programmed strip transects over 17 weeks from June to September 2023, totalling a survey area of 3.37 km^2, thereby providing porpoise monitoring data covering 89% of a special area of conservation for the species. All UAV video data were manually reviewed by a primary observer, and 25% of the UAV footage was also reviewed by a second, unexperienced observer to identify observer bias and learning effects. No significant observer bias or learning effect was found, but increased sea state affected porpoise density negatively. From the monitoring data, we were able to calculate relative density estimates, identify small-scale spatio-temporal differences and detect negative effects of recreational boat activity on porpoise presence. We further demonstrate that within this restricted area, porpoises are found in higher relative densities outside a designated conservation area, compared to within the conservation area, providing important knowledge to guide fine-scale local conservation actions. We highlight advantages and areas of improvement of UAV transect monitoring as an accessible, versatile and adaptable method to survey marine megafauna in spatially restricted specific areas of interest. We conclude that this method constitutes a promising and valuable tool for wildlife monitoring, especially as it can be easily adapted and modified for specific contexts and species.

Introduction

The harbour porpoise (*Phocoena phocoena*) is a small odontocete cetacean that feeds on a range of benthic and pelagic fish (Heße et al., 2025; Stedt et al., 2025). It has a high energy demand (Rojano-Doñate et al., 2018), and prey is considered one of the most important drivers of harbour porpoise distributions (Sveegaard, Andreasen, et al., 2012; Sveegaard, Nabe-Nielsen, et al., 2012; Gilles et al., 2016). Porpoise densities vary significantly based on regional and local conditions (Gilles et al., 2023), which result in spatial and temporal density gradients on both large (100 km to kilometres) and fine (kilometres to 100 m) scales (Gilles et al., 2009; Hammond et al., 2002; Stedt et al., 2023; Sveegaard, Andreasen, et al., 2012; Sveegaard, Nabe-Nielsen, et al., 2012; Thomsen et al., 2007; Williamson et al., 2021). Like many other coastal marine species, the harbour porpoise is threatened by anthropogenic activities (e.g. Brownell et al., 2019; Owen et al., 2024; Wisniewska et al., 2018) and protected by international

© 2025 The Author(s). *Remote Sensing in Ecology and Conservation* published by John Wiley & Sons Ltd on behalf of Zoological Society of London. This is an open access article under the terms of the Creative Commons Attribution-NonCommercial License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited and is not used for commercial purposes.

and national legislation, such as the European Union Habitats Directive (92/43/EEG, n.d.), the Convention of the Conservation of Migratory Species (CMS, 2024) and the Marine Strategy Framework Directive (2008/56/EC, n.d.).

In northern Europe, harbour porpoise population monitoring is conducted regularly through standardized inventories and provides population density estimates, large-scale distribution maps and identification of high-density areas to facilitate conservation (e.g. Gilles et al., 2023; Unger et al., 2021). Population density estimates for harbour porpoises and other cetaceans are typically obtained using distance sampling from ship- or aerial-based surveys or passive acoustic monitoring (PAM) (e.g. Gilles et al., 2023; Viquerat et al., 2014; Williamson et al., 2016). In ship-based surveys, observers scan the water surface from a ship that follows a transect to count animals (e.g. Viquerat et al., 2014). Aerial surveys use small, manned planes to scan the water surface (Jefferson et al., 2016; Scheidat et al., 2012). Seldomly, this also involves photos or videos being taken during surveys in more recent years (Williamson et al., 2016). Availability bias denotes conditions when animals are present in an area but not available for detection by an observer, due to, for example, obstruction from view or lack of vocalizations, whereas in perception bias, animals are available for detection (e.g. visible or vocalising) but still missed by the observer (Fuentes et al., 2015). Density estimates from PAM are based on long-term acoustic data loggers that detect the echolocation and communication sounds exhibited by cetaceans (Marques et al., 2013). PAM allows for standardized data collection with lower effort and is largely independent of weather, environmental conditions and time of the day (Amundin et al., 2022). However, challenges include masking of cetacean signals by other underwater sounds, analysis of large quantities of data, varying detection range limits, false positives, behavioural variability in detection probability (Macaulay et al., 2023) and obtaining precise data on abundances and calf-to-adult ratios (Amundin et al., 2022).

More detailed studies of harbour porpoise small-scale distribution and habitat use within identified important areas or other restricted areas of specific interest have previously primarily relied on PAM (Benjamins et al., 2017; Stedt et al., 2023) or traditional visual surveys using, for example, a theodolite to track porpoise movements (Brandt et al., 2013; Dinkel et al., 2025; Jones et al., 2014). However, new methodologies for population monitoring and small-scale habitat use of porpoises and other elusive marine megafauna should be explored as they can increase our understanding of a species' ecology and support management actions for threatened populations.

Unmanned aerial vehicles (UAVs) or 'drones' are small airborne vehicles that are operated remotely by one or two persons, usually from land or boat with vertical take-off and landing ability in multirotor UAVs (Colefax et al., 2018). Multirotor UAVs can act as inexpensive, low-risk, accessible alternatives to manned aircraft surveys, that offer high-resolution data with fewer logistical and legal constraints, especially at fine and small scales and in remote areas and habitats, which may not always be accessible via aircraft or ship (Anderson et al., 2025; Colefax et al., 2018; Sweeney et al., 2016). They can assess observer bias, counting error, availability and perception bias and species misidentification by thorough review of high-resolution video or image (Fettermann et al., 2022; Hodgson et al., 2013, 2017; Johnston et al., 2017). While the study design and data review may be conducted by experts or trained staff, digital aerial surveys do not always require the presence of experts during data collection in the field and as little as one person can be sufficient to pilot a UAV.

In recent years, UAVs have advanced in technology and become more accessible to study the movements, morphometry and behaviour of marine mammals (e.g. Brennecke et al., 2022; de Oliveira et al., 2023; Frouin-Mouy et al., 2020; Koski et al., 2015; Stedt et al., 2024) but also sharks (Butcher et al., 2021) and turtles (Rees et al., 2018). Most previous studies have focused on assessing detection probability compared to other visual surveying methods and in response to environmental variables and camera settings (Angliss et al., 2018; Aniceto et al., 2018; Hodgson et al., 2013, 2017; Koski et al., 2009). However, with relatively small effort and low cost, UAVs provide data quality as high as, or higher than, other sampling methods for monitoring (Aniceto et al., 2018; Fettermann et al., 2022; Hodgson et al., 2017; Johnston et al., 2017). UAV data allow for the study of behaviour and morphology simultaneously with density and abundance assessments, which have the potential to provide novel insights into marine mammal behaviour, social structure, morphometry and life history (de Oliveira et al., 2023; Hodgson et al., 2017). To date, we are not aware of any published studies investigating harbour porpoise density and distribution using UAVs.

The aim of this study was to (1) test programmed UAV transect monitoring as a novel method for standardized collection of UAV observations of harbour porpoises, (2) investigate method applicability by studies of the spatio-temporal distribution, potential drivers of observed differences and density of harbour porpoises in a known high-density area for porpoises and (3) evaluate digital aerial surveys for use in future monitoring of elusive marine wildlife in specific areas of interest.

© 2025 The Author(s). *Remote Sensing in Ecology and Conservation* published by John Wiley & Sons Ltd on behalf of Zoological Society of London.

20563485, 0, Downloaded from https://zslpublications.onlinelibrary.wiley.com/doi/10.1002/rse2.70043 by NICE, National Institute for Health and Care Excellence, Wiley Online Library on [25/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Materials and Methods

Study area

Kullaberg is an accentuated mountain ridge and peninsula in north-western Scania, Sweden (Fig. 1). Frontal zones of upwelling nutrients and oxygenated waters are created at the tip of the peninsula when low salinity (10–20 ppm) brackish surface water from the Baltic Sea outflows from Öresund and encounters the high salinity (25–35 ppm) water from the Kattegat (Sveegaard, 2011; Unger et al., 2021). This results in a high marine biodiversity at Kullaberg (Pedersen, 1993; Sveegaard, 2011).

The harbour porpoise is found in high densities at Kullaberg, including mother–calf pairs with an estimated 6% calf proportion in the Belt Sea (Gilles et al., 2023; Unger et al., 2021; Viquerat et al., 2014). In 1986, a 2.01 km^2 marine reserve covering the area out to 300 m from the shore was established, and in 1996, the same area was designated as a Swedish Natura 2000 site for harbour porpoises. During the summer months, there is high recreational boat activity in the marine reserve as it is surrounded by coastal towns and harbours in the densely populated Öresund.

Data collection

The study area around the Kullaberg peninsula was surveyed during 17 weeks in summer (June–September) 2023 using a quadcopter UAV (model DJI Phantom 4 Pro, SZ DJI Technology Co., Ltd., Shenzhen, China). This UAV has a maximum speed of 72 km h^{-1} and a flight time of 20–30 min depending on wind speeds, which it resists up to 10 m s^{-1}. The camera has a field of view (FOV) of 84°.



Figure 1. Study area at the Kullaberg peninsula at the northern part of Öresund between southern Sweden and Denmark. The marine reserve and Swedish Natura 2000 site for harbour porpoises extends out to 300 m from the shore (light blue line). The extent of the total surveyed area out to 600 m from the shore is also shown (dark blue striped line), as is the reference area (top right inset) which was located c. 4.5 km east along the northern side of the peninsula. The colour-coded flight paths (1–7) illustrate the UAV transect monitoring path in seven sample areas, with non-sampled areas highlighted in white cross-strokes. Parts of flight path 5 were slightly adjusted to the boundaries of a bird-protected area (red striped line).

© 2025 The Author(s). *Remote Sensing in Ecology and Conservation* published by John Wiley & Sons Ltd on behalf of Zoological Society of London.

20563485, 0, Downloaded from https://zslpublications.onlinelibrary.wiley.com/doi/10.1002/rse2.70043 by NICE, National Institute for Health and Care Excellence, Wiley Online Library on [25/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Videos were recorded at a resolution of 4096:2160 pixels and an aspect ratio of 1.9. The remaining camera settings were set to default, to ensure consistent adaptation to lighting conditions with automatic ISO and focus. The UAV was operated using the designated remote control and an Android 4.4.0 tablet with the app 'Litchi' to pre-program strip transects. Test flights were performed using pre-programmed UAV speeds of 8, 10 and 12 m s^{-1} to evaluate porpoise detectability and maximum flight length, and 8 m s^{-1} did not guarantee completion of the pre-programmed flights on one battery pack under all relevant wind conditions. Instead, a standard flight speed of 10 m s^{-1} was chosen for the transects as it also allowed a higher detection ability of porpoises than at 12 m s^{-1}.

The study area was subdivided into seven sub-areas (hereafter referred to as 'areas'); six areas within the known high-density area for porpoises and one reference area c. 4.5 km east with similar methodological prerequisites, but with a documented lower sighting rate of porpoises (Fig. 1). This reference area number 7 was chosen to allow comparison with the areas of proposed high density around the peninsula and detection of potential large differences in relative densities from the tip of the peninsula to the east. Areas were defined based on take-off point accessibility, the ability to keep the UAV within the line of sight and a stable remote-control signal for most of the flight. For these reasons, there was a gap between areas 2 and 3, 3 and 4, and 5 and 6 (Fig. 1). All flights were performed between 9 AM and 4 PM local time (UTC + 2) in varying start-to-end order.

Data collection was performed by the same person throughout the entire study and at wind speeds ≤8 m s^{-1} to ensure a calm and consistent sea state and UAV safety (Ayres et al., 2021). During pre-programmed flights, the camera angle was set to 90° (pointing straight down) as a standard, but it was adjusted manually if required. Each flight covered one of the areas fully on a single battery pack. All seven pre-programmed flights were performed twice per week over 17 weeks, except for the reference area, which was sampled once per week.

The aim was a UAV altitude of 66 m, which gives 5% overlap at 100 m distance between transects in case of wind affecting the UAV and small imprecisions in the flight path, altitude and camera angle, and transects were laid against the depth gradient perpendicular to the shore (Buckland et al., 2001; Hodgson et al., 2013). For the purpose of estimating density, the overlap between transects was truncated as we had insufficient data and detections of animals in the overlap. Instead of investigating the effects of glare from this data according to Hodgson et al. (2013), we modelled cloud cover as a proxy for glare. We assume this had no notable effect on our data. The FOV of the video was calculated according to Equation (1) to determine the total unique area sampled per flight, which depends on the UAV altitude and number of transects in the area. The cameras' FOV was 55 m × 105 m. Flights at the pre-programmed altitude of 66 m resulted in a 2.6 cm^2 pixel size and 1.6 cm × 1.6 cm ground sampling distance. However, due to the UAV barometer precision there was a ±10 m margin of error, which was accounted for using a parametric bootstrap.

$$\mathrm{FOV} = \frac{2 \times a \times h \times tantan\,(r/2)}{\sqrt{1+a^2}} \times \frac{2 \times h \times tantan\,(r/2)}{\sqrt{1+a^2}} \quad (1)$$

FOV (size of the area captured in one frame) calculation where a is the aspect ratio, h the UAV altitude and r the camera's FOV angle in radians.

Each area and pre-programmed flight consisted of nine or 10 parallel strip transects, each at 100 m from the next transect and 500 m long perpendicular to the shore (Fig. 1), with a total transect strip length of 5 300–6 400 m per flight. Due to restrictions of a bird protective area in area 5, four of the transects in this area were altered from the standard to start an additional 50 m further away from the shore (Fig. 1). We categorized sampling areas into South (areas 1 and 2 on the southern side of the peninsula), Tip (areas 3 and 4 at the tip of the peninsula) and North (areas 5 and 6 on the northern side of the peninsula) (Table 1), due to previously observed differences in relative porpoise activity between these sites (Stedt et al., 2023). The reference area, area 7, was not included in this comparison. Sampling areas in random order was logistically impossible due to the distances between take-off points and time constraints. Instead, we systematically varied the sampling order by starting in a

Table 1. Details of each sampling area.

Sampling area	Size estimate lower km^2	Size estimate upper km^2	No of transects	Side
Area 1	0.44	0.48	9	South
Area 2	0.44	0.48	9	South
Area 3	0.44	0.48	9	Tip
Area 4	0.49	0.53	10	Tip
Area 5	0.49	0.53	10	North
Area 6	0.49	0.53	10	North
Area 7 (reference)	0.44	0.48	9	Reference

Sizes were calculated based on the UAV path, thereby representing the total surface area sampled during each flight in each of the areas, including transects perpendicular to the shore and 100 m paths between transects. 'Side' refers to the sampling area position around the Kullaberg peninsula.

© 2025 The Author(s). *Remote Sensing in Ecology and Conservation* published by John Wiley & Sons Ltd on behalf of Zoological Society of London.

20563485, 0, Downloaded from https://zslpublications.onlinelibrary.wiley.com/doi/10.1002/rse2.70043 by NICE, National Institute for Health and Care Excellence, Wiley Online Library on [25/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

different sampling area each day, thereby achieving even temporal coverage of the areas.

Although the pre-programmed UAV speed was set to $10\,m\,s^{-1}$ throughout the study, local wind conditions affected the acceleration and turning speed of the UAV, which caused some differences in flight length, with 93.1% of flights lasting 10–13 min. After each flight, local weather data were obtained from a public weather service called 'Kustväder' supplied with data from the Norwegian Meteorological Institute and NRK (Sjöräddningssällskapet, Swedish Sea Rescue Society). Sea state was assessed using the Beaufort Sea State Scale (Table S1). Cloud cover was graded on a scale from 1 to 4 according to the approximate percentage of the sky that was visible (Table S2). The number of boats present was counted for each flight by observing any boats passing through the sampling area and categorized into sailing boats, motor boats, jet skis and small unmotorized boats (e.g. kayaks). This information was always recorded by the same UAV pilot in the field by visual observation right after the flight had been completed.

Video analysis

Videos were analysed by a single person (the same person who collected the data in the field) on a 13.3-inch 2020 MacBook Air Retina Display at 400 nits brightness in a dark room to minimize light reflections on the screen. A second independent observer without experience in detecting porpoises was briefed to recognize porpoises and seals (*Phoca vitulina* or *Halichoerus grypus*) in the footage and reviewed 25% of the footage in random order. This was done to compare detections between the two independent observers and test for significant differences, which would indicate an observer effect. Further, these detections were examined to test for the learning effect, that is, an inexperienced observer improving in their ability to detect animals over time. Group size was defined as the total number of porpoises detected in one frame of video. The software Video GeoTagger (Remote GeoSystems, USA) was used to find the GPS position the UAV was at during each observation. ArcGIS ver. 10.8.2 (Esri, USA) was used to record the distance from the shore. Each video was reviewed twice at normal speed. The final dataset is freely available through the online data repository Mendeley (Hartmann, 2025).

Statistical analysis

Estimating and modelling porpoise density

All statistical and exploratory analyses were conducted in R studio version 4.1.1. Density was estimated based on the primary observer's detections only. A basic framework was used to estimate animal density in each area according to Equation (2) (Buckland et al., 2001), assuming strip transects with uniform detectability and not accounting for the probability of surveying (the probability of a transect being placed in the area) and detecting animals (the probability of observing and not missing detections due to, e.g. perception or availability bias) (Hammond et al., 2021). We assume no false positive detections (objects falsely identified as porpoises), which is valid as in the study area it is very rare to encounter other cetacean species, and porpoises had a stereotypical appearance in the footage and distinct movements making them easily identifiable. It is reasonable to assume our estimates are representative of the study area as 89% of the study area was sampled during each sampling day. Each individual flight performed in an area was treated as a separate sample (sampling unit). Even though it is possible for harbour porpoises to swim between areas within one sampling day, the independence assumption was relaxed as the main interest was relative animal distribution patterns in terms of frequently used areas by porpoises and the associated predictors, rather than obtaining absolute abundance and density estimates.

$$\widehat{D} = \frac{\widehat{N}}{A} \tag{2}$$

Basic equation to calculate density in a sampled area. $\widehat{N}$ delineates porpoise detection counts, that is, number of animals in an area size A in a flight (Buckland et al., 2001). Area size error due to variable UAV altitude has been propagated via bootstrap to produce a distribution of density estimates for each area to give mean estimates of density and 95% CIs.

To obtain the range estimate of area size of $0.44–0.53\,km^2$ based on UAV altitude error, a normal distribution with a mean of 66 m (our altitude setting) and standard deviation of 10 m (UAV altitude error) was assumed. A parametric bootstrap of 10 000 iterations was performed on this distribution to obtain 95% confidence intervals (CIs) on area estimates and means based on bootstrapped altitudes used to calculate area size according to Equation (1). For areas 1, 2, 3 and 7 with 9 transects, the range was estimated as $0.44–0.48\,km^2$ with a mean of $0.46\,km^2$. For areas 4, 5 and 6 with 10 transects, the range was $0.49–0.53\,km^2$ with a mean of $0.51\,km^2$. This error was then propagated into our density estimate and 95% CIs by simulating porpoise counts for each area based on the distribution of our count data and by simulating area size according to UAV altitude distribution through 10 000 bootstrap iterations.

© 2025 The Author(s). *Remote Sensing in Ecology and Conservation* published by John Wiley & Sons Ltd on behalf of Zoological Society of London.

20563485, 0, Downloaded from https://zslpublications.onlinelibrary.wiley.com/doi/10.1002/rse2.70043 by NICE, National Institute for Health and Care Excellence, Wiley Online Library on [25/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Using the package 'lme4' (Bates et al., 2015), a mixed effect generalized linear model (GLMM) was fitted to the data with the number of porpoises detected per flight as the response variable. The model assumed a Poisson distribution with a logarithmic link function, since the response variable was a count and over-dispersion tests were negative. Exploratory analysis and variance inflation factor (VIF) were used to rule out multi-collinearity between predictor variables. The seven areas were treated as independent units and entered as random effects into the model to account for autocorrelation between flights. A backwards selection approach was used and the model with the lowest AIC was selected until the AIC did not change by more than 2 (Burnham & Anderson, 2002), which resulted in temperature and wind speed being eliminated from the model. In the final model, an extreme outlier was excluded due to atypical sampling conditions of sudden rain that resulted in early termination of the flight.

When there was more than one harbour porpoise observation during a flight, group sizes and distances between groups were examined manually to check if the same group was resampled and any duplicates were excluded. Due to the dependent relationship between the time porpoises spent at the water surface for detection and detectability, only relative densities and distribution patterns over time were investigated and therefore perception and availability bias were unaccounted for. Obtaining estimates of time spent at the surface across our study area would be possible given the detailed UAV footage, but it was outside the scope of this work. Variability in the size of each area that was sampled was accounted for in our area size estimates and by using the logarithm of the area size of each flight as an offset in the models.

A drop-in-deviance test using ANOVA was performed on the final model and on the same model with sampling area added as a fixed effect.

Learning effect and observer bias

For the second observer who reviewed videos in a random order, a Spearman rank correlation between video review order and porpoise counts was used to evaluate a potential learning effect. A generalized linear model (logarithmic link function, Poisson distribution) was fitted to test for correlation of video order and porpoise counts, with the second observer's porpoise counts as the response variable, the logarithm of area size as offset, and video review order and the first observer's porpoise counts as predictors to account for 'true' variation in porpoises available for detection in the videos. This way unnecessary model complexity was avoided by capturing underlying patterns in the data using the first observer's observations instead of environmental covariates. The Phi coefficient and paired Wilcoxon signed-rank test statistic were calculated to examine the association between the two different observers when treating porpoise observations as binary data in a contingency table and as paired observations.

Results

Porpoise density and distribution

Flights and descriptive statistics

From 6 June 2023 until 27 September 2023, 188 UAV flights were performed over 17 weeks and 30 sampling days (Table 2). The seven areas covered a total of 3.37 km^2 (CI 3.23–3.51 km^2), thereby allowing data collection from 89% of the total area of the marine reserve (stretching out to 300 m from the shore) plus an additional area outside the reserve (200 m from the outer boundary of the marine reserve) for each transect. In total, 78 harbour porpoise *Phocoena phocoena* individuals were observed by UAV over the study period, with observations in 19% of flights (Table 2). The mean number of harbour porpoises per flight was 0.41 and the mean density was 0.95 ind km^{-2} (CI 0.51–1.40). Out of the 261 boats recorded, 146 were motorised with a mean of 0.78 motorboats per flight (Table 2).

Five calves were identified in the UAV data, representing 6% of all identified porpoises. All calf observations were made during four sampling days between 17 August and 26 September. Seven seals (*Phoca vitulina* or *Halichoerus grypus*) were observed across all 4 months (Table 2).

No significant correlation between any of the explanatory variables was found. The Pearson correlation coefficient was <0.7 for all pairwise correlations of Beaufort Sea state, temperature, wind speed, cloud cover, motor boats and day and VIF factors in the GLMM ranged from 1.04 to 1.15 (Table 3).

Temporal distribution

There was a seasonal effect, with the number of detected porpoises increasing in August and September (est. $= 5.00$, $P = 0.0071$) (Table 3, Fig. 2). Diel differences in the number of detections were found, with more porpoises detected the earlier in the day an area was sampled (est. $= -0.22$, $P = 0.026$) (Table 3).

Spatial distribution

Harbour porpoise density ranged from 0 to 28.27 ind km^{-2}. The highest mean density was found in areas 5

© 2025 The Author(s). *Remote Sensing in Ecology and Conservation* published by John Wiley & Sons Ltd on behalf of Zoological Society of London.

20563485, 0, Downloaded from https://zslpublications.onlinelibrary.wiley.com/doi/10.1002/rse2.70043 by NICE, National Institute for Health and Care Excellence, Wiley Online Library on [25/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 2. Descriptive statistics of total and mean number of harbour porpoises and motor boats, and total number of seals, observed per sampling area and flight as well as summarized results for the complete dataset.

Sampling area	Total number of porpoises	Mean number of porpoises per flight	Total flights	% of flights with porpoises	Total number of motor boats	Mean number of motor boats	Total number of seals
Area 1	3	0.10	29	6.90%	18	0.62	2
Area 2	14	0.48	29	13.79%	25	0.86	0
Area 3	9	0.31	29	20.69%	24	0.83	1
Area 4	10	0.36	29	25.00%	25	0.89	1
Area 5	21	0.72	29	27.59%	27	0.93	2
Area 6	17	0.59	28	20.69%	18	0.59	1
Area 7	4	0.27	15	13.33%	9	0.60	0
Total	78	0.41	188	18.60%	146	0.78	7

Table 3. Test statistics for individual predictors of the final GLMM (AIC = 314.40) for harbour porpoise counts offset by sampling area size, including slope estimate, standard error, *z* value, Wald statistic (*P*-value), confidence intervals (CI) and variance inflation factor (VIF) for sampling day, area, hour of the day, sea state, cloud cover and number of motor boats.

	Estimate	Std. error	*z* value	*P*-value	2.5% CI	97.5% CI	VIF
(Intercept)	1.7	1.4	−1.3	0.21	−0.91	4.4	
Day	5.00	1.86	2.69	0.0071**	1.5	8.8	1.05
Time of day	−0.22	0.10	−2.22	0.026*	−0.43	−0.033	1.04
Beaufort Sea state	−0.71	0.20	−3.50	0.00047***	−1.1	−0.32	1.15
Cloud cover	−0.10	0.15	−0.65	0.51	−0.40	0.20	1.10
Motor boats	−0.28	0.13	−2.11	0.035*	−0.56	−0.047	1.07

Signif. codes: '***' 0.001, '**' 0.01, '*' 0.05.

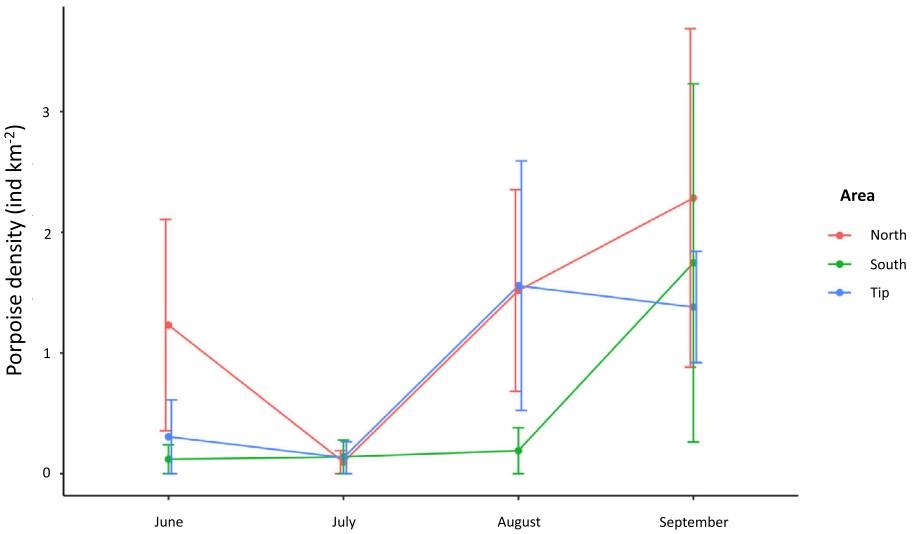


Figure 2. Porpoise density in ind km^{-2} for each sampling month from June to September categorized by sampling area position around the Kullaberg peninsula. Standard error bars are based on the variability of density within each month, which are the estimates that also include propagated error based on UAV altitude error. 44 flights were performed in June, 42 in July, 58 in August and 44 in September.

and 6 with 1.4 and 1.2 ind km^{-2} (CI 0.86–2.1 and 0.61–1.7) respectively. The lowest mean density was found in area 1 with 0.22 ind km^{-2} (CI 0–0.52). The highest densities were observed in areas 4, 5 and 6 (Table 4). Areas 1, 3 and 7 (the reference area) showed consistently lower densities (Table 4). Especially areas 5 and 6 located

© 2025 The Author(s). *Remote Sensing in Ecology and Conservation* published by John Wiley & Sons Ltd on behalf of Zoological Society of London.

20563485, 0, Downloaded from https://zslpublications.onlinelibrary.wiley.com/doi/10.1002/rse2.70043 by NICE, National Institute for Health and Care Excellence, Wiley Online Library on [25/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 4. Random effects of the sampling areas calculated for the GLMM with the response variable of porpoise counts offset by sampling area size with predictor variables sampling day, time of day, sea state, cloud cover and number of motor boats.

	Intercept	Mean density ind km^{-2}	Density CI
Area 1	−0.58	0.22	0–0.52
Area 2	−0.38	0.31	0.75–0.63
Area 3	0.0056	0.67	0.29–1.1
Area 4	0.29	0.70	0.28–1.2
Area 5	0.65	1.4	0.86–2.1
Area 6	0.54	1.2	0.61–1.7
Area 7	−0.32	0.58	0.14–1.2
Total		0.95	0.51–1.40

Intercepts and density estimates are given excluding an identified outlier. CI represents 95% confidence intervals. Mean densities, density CI's and density ranges are given for every area.

on the northern side showed both a high density of observations and a high number of individuals per observation, with a mean group size of 1.75 individuals in area 5 (CI 1.14–2.36). The overall mean group size was 1.56 individuals (CI 1.24–1.88, range 1–6). At the tip of the peninsula (areas 3 and 4), observations were frequent but group sizes were small. Areas 4, 5 and 6 had significantly higher densities compared to areas 1, 2 and 7 (Table 4). A drop-in-deviance test using ANOVA showed a significant effect of sampling area when added to the model ($P = 0.016$).

Porpoises were detected at a mean distance of 369 m (CI 324–414) from the shore, although they were found as close as 39 m and as far as 663 m away (Table S3). A majority (64%) of observations were made outside the 300 m boundary of the marine reserve (Fig. 3), although all transects were laid out half inside and half outside the reserve thus resulting in an equal sampling effort across the areas. This design did, however, result in a gap at the tip of the peninsula, outside the marine reserve, that was not sampled (Fig. 3).

Effects of environmental and anthropogenic factors

As Beaufort Sea state increased, porpoise density decreased significantly (est. = −0.71, $P = 0.00047$) (Fig. 4). There was also a negative relationship between porpoise density and the number of motor boats (est. = −0.28, $P = 0.035$, Table 3).

Observer bias and learning effect

Forty-eight flights were reviewed by a second independent observer. In 42 out of the 48 flights (88%), the detection count did not differ between observers (Fig. 5, Table 5). In six flights (12%), one observer missed an observation made by the other (Fig. 5, Table 5). There was no difference between the observers in the number of missed observations as both missed three detections each (Table 5). The Phi coefficient was 0.621, indicating a positive relationship between the observers. The non-significant results of the paired Wilcoxon signed-rank test further support alignment between the results of the two observers ($V = 26.5$, $P = 0.6675$).

There was no evidence for a learning effect, as there was no systematic relationship between video review order and porpoise counts in the GLM for the second observer (est. = −0.021, $P = 0.16$, Table S4). This is consistent with the Spearman rank correlation between video review order and porpoise counts, which was non-significant ($S = 19112$, $P = 0.80$).

Discussion

Porpoise density and potential drivers

Porpoise density and distribution

With an estimated density of 0.95 ind km^{-2} from UAV transect monitoring, the area immediately surrounding the Kullaberg peninsula in northern Öresund is to be considered a high-density area for harbour porpoises belonging to the Belt Sea population of porpoises. Compared to the overall population estimates for the larger areas previously surveyed in large-scale inventories in the Belt and North Sea (Gilles et al., 2023; Teilmann et al., 2008; Unger et al., 2021), the higher estimated densities at Kullaberg in the current study indicate that this area is a locally important habitat for the Belt Sea population. It should, however, be noted during these comparisons that these large-scale inventories report absolute abundances and densities rather than relative densities which are reported in this current study. Many marine mammals, porpoises included, aggregate in restricted areas in time and space in correspondence to presumed availability of the resources they depend on for survival (e.g. Jones et al., 2014; Stedt et al., 2023; Sveegaard, Nabe-Nielsen, et al., 2012). Northern Öresund and Kullaberg have previously been identified as a porpoise 'hotspot', possibly due to high prey biodiversity and abundance shaped by currents, nutrient flow and strong easterly winds in spring (Stedt et al., 2023; Sveegaard, 2011; Teilmann et al., 2008, 2022) and our results corroborate this notion. As the Belt Sea population of porpoises reproduce during late summer (Lockyer & Kinze, 2003), the high density of porpoises in the area during August and September may also be because

© 2025 The Author(s). *Remote Sensing in Ecology and Conservation* published by John Wiley & Sons Ltd on behalf of Zoological Society of London.

20563485, 0, Downloaded from https://zslpublications.onlinelibrary.wiley.com/doi/10.1002/rse2.70043 by NICE, National Institute for Health and Care Excellence, Wiley Online Library on [25/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

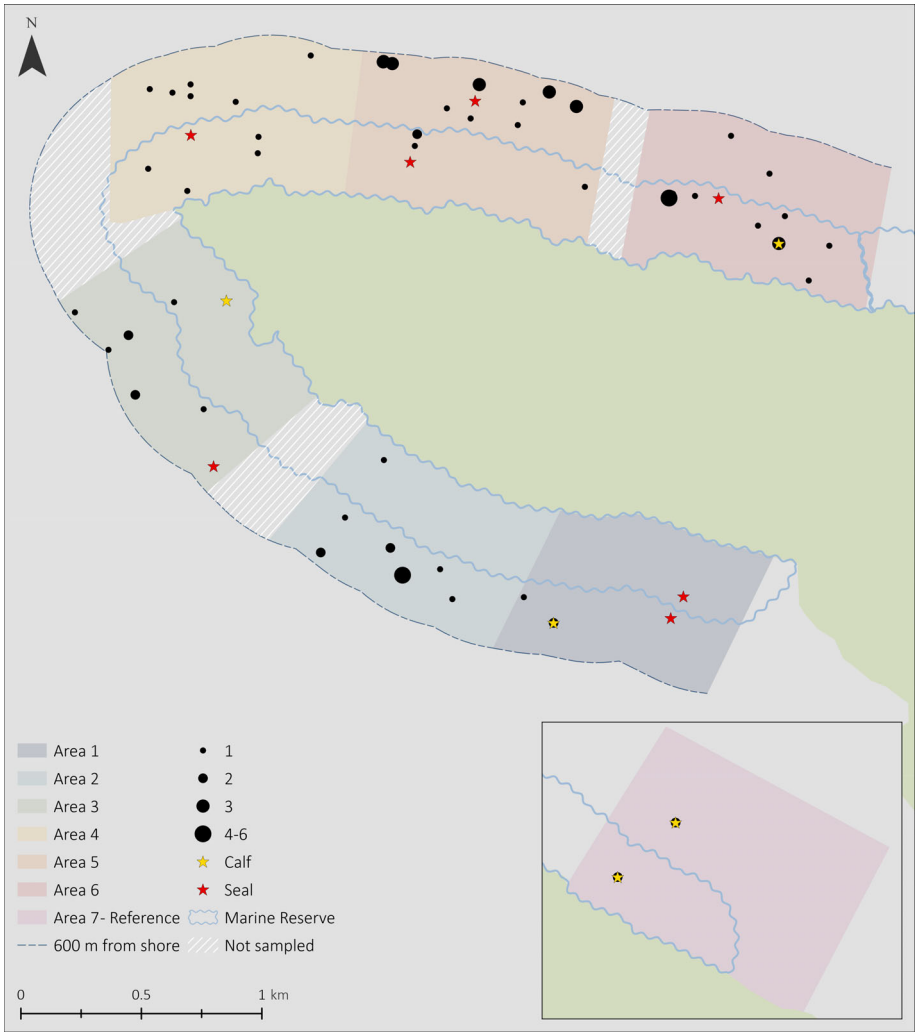


Figure 3. Porpoise (black marks) and seal (red star) observations by UAV transect monitoring in the surveyed area within 600 m from the shore for the complete study period, with each of the seven sub-areas monitored by respective flight paths indicated by area-specific colour coding. Non-sampled areas are highlighted in white cross-strokes. The size of the black marks indicates the number of porpoises (group size) for that observation, with calves indicated separately and overlaid (yellow stars).

individual porpoises move into a specific reproductive area to facilitate mating. This is supported by opportunistic observations made during data collection of porpoises exhibiting mating behaviours, such as 'acrobatic jumps' out of the water (Norman et al., 2018), and similar records from late summer in previous years from the same area (Stedt, unpubl. data; Webber et al., 2023).

The observed spatial patterns, with higher relative densities at the tip of the peninsula and lower relative densities on the southern side and further east align with previous local studies using PAM (Stedt et al., 2023). Porpoises likely respond to differences in prey abundance and diversity (Gallagher et al., 2021) between closely located areas as they are known to forage continuously due to high relative energy requirements (Wisniewska et al., 2016). Further work is required to assess the local small-scale habitat and prey availability differences that may act as underlying drivers of porpoise distribution, density and activity between the northern side, the southern side and the tip of the Kullaberg peninsula. Also, in PAM, detection probability has been shown to depend on behavioural state (Macaulay et al., 2023), and dive behaviour could also bias the probability of visual or UAV detection, so the mechanism behind apparent changes in relative density needs to be investigated to make further inferences based on relative density.

Stedt et al. (2023) found porpoise activity to be highest in the area from September to December, which is

© 2025 The Author(s). *Remote Sensing in Ecology and Conservation* published by John Wiley & Sons Ltd on behalf of Zoological Society of London.

20563485, 0, Downloaded from https://zslpublications.onlinelibrary.wiley.com/doi/10.1002/rse2.70043 by NICE, National Institute for Health and Care Excellence, Wiley Online Library on [25/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

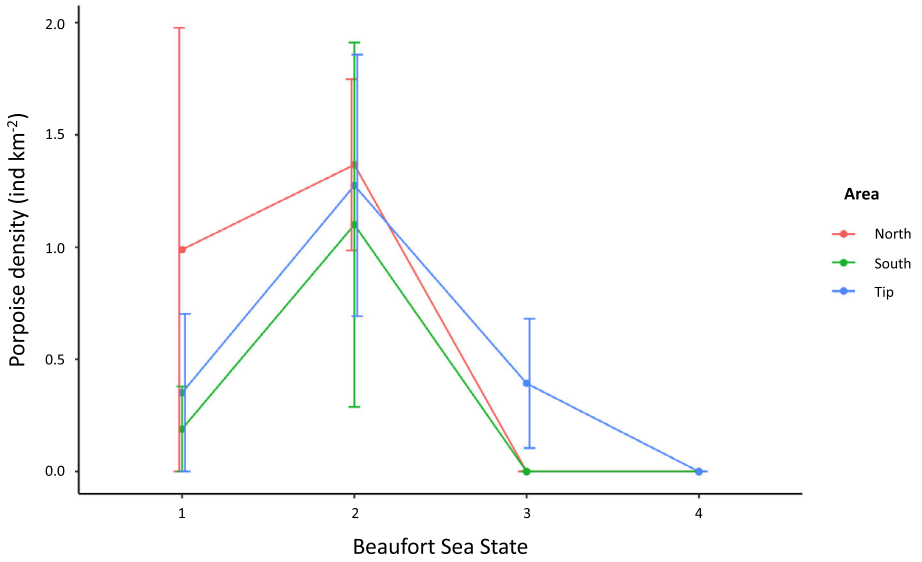


Figure 4. Porpoise density in ind km^{-2} over Beaufort Sea state categorized by sampling area position around the Kullaberg peninsula and standard error bars with propagated error based on UAV altitude.

consistent with the observed increase in density in August and September. We also demonstrate that the majority (64%) of detections were made outside the marine reserve boundary at 300 m from the shore, which has management implications as porpoises are more likely to be exposed to disturbances in the area outside the reserve, for example, via unregulated recreational motorboat activity.

Calves

Calf proportions at harbour porpoise calving and nursing grounds have been reported to be up to 18% (Sonntag et al., 1999), with proportions at Kullaberg previously reported at 10% (Stedt, unpubl. data). The percentage of calves (6%) found in this study is the same as the overall average calf proportion found for the Belt population (Unger et al., 2021; Viquerat et al., 2014), indicating no special importance of the Kullaberg area for mother–calf pairs. It is also possible that newborn and very young calves were too small for detection by the UAV monitoring methodology in June–July and that reliable detection of calves using the current monitoring methodology is only possible during a narrow and restricted time window as porpoise calves rapidly grow to sizes where they might be misclassified as adults. Calves grow to 84% of their adult size within the first year (Stepien et al., 2023). Using Equation (3), as described in Stepien et al. (2023), and assuming birth around May–June gives an approximate body length of 97 cm in August (3 months old) and 101 cm in September (4 months old). The proportion of calves found here may also be an underestimate as calves spend a large proportion of their time swimming underneath the mother in infant position or in echelon position (Clausen et al., 2011; Hamel et al., 2025) which decreases their detectability in UAV footage. These potential calf detection biases are important to consider and explore in future studies, which could be achieved via UAV data to give morphometric measurements or dedicated focal follows. The aggregation of calf detections over a short period of time may also hint at potential temporal autocorrelation, due to possible re-sampling of mother-calf pairs with high site fidelity (Camphuysen & Krop, 2011; Hamel et al., 2025), that increased calf counts during this specific period.

$$\text{Length} = 1.4\exp(-0.542(\exp(-0.119\text{AgeMonth}))) \quad (3)$$

Gompertz growth curve to describe the relationship between porpoise length and age in months, developed by Stepien et al. (2023).

Recreational motorboats

As the number of recreational motorboats did not vary noticeably between areas, their significant negative effect on porpoise density suggests that porpoises move between areas in response to prevailing boat presence. The overall co-occurrence of high porpoise densities and large numbers of recreational motorboats in the Kullaberg area suggests that porpoises do not necessarily avoid regions with high levels of human disturbance altogether, but perhaps

© 2025 The Author(s). *Remote Sensing in Ecology and Conservation* published by John Wiley & Sons Ltd on behalf of Zoological Society of London.

20563485, 0, Downloaded from https://zslpublications.onlinelibrary.wiley.com/doi/10.1002/rse2.70043 by NICE, National Institute for Health and Care Excellence, Wiley Online Library on [25/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

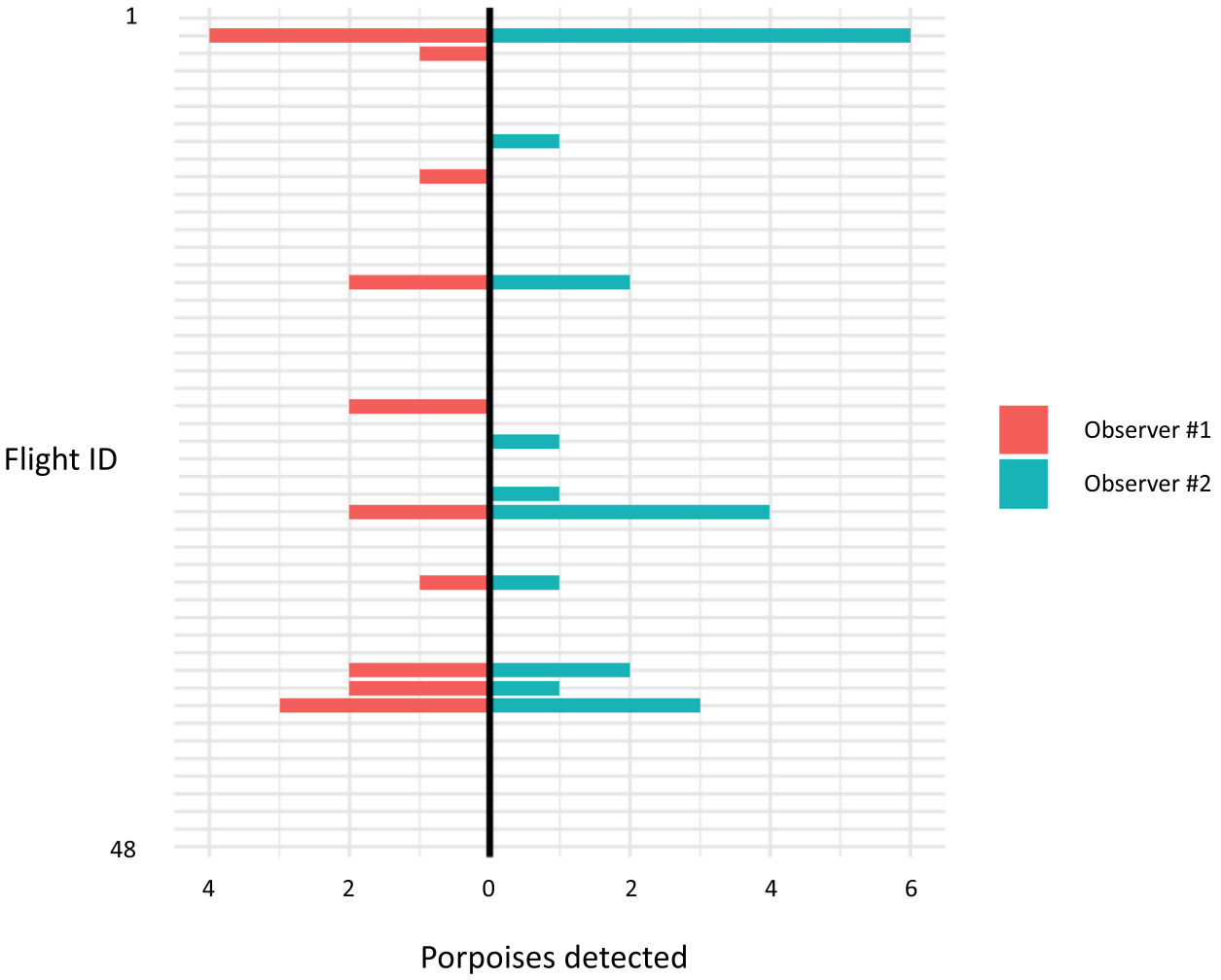


Figure 5. Comparison of porpoise observations in 48 different flight videos between two independent observers. The x-axis shows the number of porpoises detected in a flight that was reviewed by both observers, and the y-axis shows flight ID 1–48.

Table 5. Contingency table showing how many times each observer detected harbour porpoises (HP) in the 48 randomly chosen flights that were reviewed by the primary observer (observer 1) and a second independent observer (observer 2).

	HP detected by observer 1	HP not detected by observer 1
HP detected by observer 2	7 (15%)	3 (6%)
HP not detected by observer 2	3 (6%)	35 (73%)

rather adjust their behaviour to allow continued use of important habitats. During this study, we made several observations of high-speed motorboats directly approaching and driving through groups of porpoises, causing a clear response and behavioural change, with porpoises often disappearing from the surface for prolonged periods. This supports the idea that motorboat presence disturbs porpoises, even at distances of several 100 m, causing them to change their behaviour and expend extra energy (Frankish et al., 2023; Hao et al., 2024; Till et al., 2025). Underwater noise caused by motorised boats is known to disturb porpoises (Dyndo et al., 2015; Hao et al., 2024; Hao & Nabe-Nielsen, 2023). Such noise may disrupt foraging behaviour through masking (Wisniewska et al., 2018) or disrupt navigation and communication (Frankish et al., 2023), causing disorientation and possibly disruption of social behaviours that may be crucial for reproductive success and social structure. Auditory masking by motorised vessels has been demonstrated in porpoises (Hermannsen et al., 2025), and disorientation may increase the risk of entanglement in gill nets (Nielsen et al., 2012). In our study, we assume that the combined physical and noise effect that is seen in motorboats likely has the strongest impact and that the effect of, for

© 2025 The Author(s). *Remote Sensing in Ecology and Conservation* published by John Wiley & Sons Ltd on behalf of Zoological Society of London.

20563485, 0, Downloaded from https://zslpublications.onlinelibrary.wiley.com/doi/10.1002/rse2.70043 by NICE, National Institute for Health and Care Excellence, Wiley Online Library on [25/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

example, sailing boats, that do not generate motor noise during active sailing, is weaker. We do not explore this further in our data, but this is a research area that warrants further investigation.

Benefits and limitations of UAV transect monitoring

Benefits of UAV transect monitoring

The here proposed UAV transect monitoring method provides high-spatio-temporal-resolution data on small-scale habitat use that is more affordable, accessible and less invasive than ship or aircraft-based monitoring data (Verfuss et al., 2019). UAV video data can be copied, shared and analysed independently and repeatedly in space and time, allowing for multiple independent reviews of the original observational data by many independent observers, where ship- and manned-aircraft surveys typically only have space to allow for a limited number of observers during surveys (Verfuss et al., 2019). Further analysis and exploration of the video data can thereby provide further insights into, for example, perception bias and observer effects. New detections between observers are expected to plateau as multiple independent reviews are performed, representing the number of total animals available for detection. Subsequently, perception bias can be quantified using the proportion of available animals detected by independent observers.

High-resolution UAV video data enable elaborate and detailed studies of animal behaviour, movement patterns and morphometrics with high temporal coverage alongside animal density and distribution analyses (Hodgson et al., 2017). Replicating the effort of our survey in boat or manned-aircraft surveys would be notably more expensive, difficult to implement and invasive (Verfuss et al., 2019).

Observer fatigue, a known issue during aerial or boat surveys (Ransom, 2012), can be mitigated, as observers can pause and take breaks. Replaying sequences and zooming in to obtain accurate counts may increase precision compared to lower resolution surveys. Our data could be used to explore these effects in more detail by, for example, modelling perception bias depending on different modes of video review and frequency of breaks.

For studies of habitat use and monitoring within restricted areas, the UAV transect monitoring methodology presents several benefits compared to traditional monitoring methods while most limitations described for UAVs also apply to traditional visual survey methods (Oliveira-Rodrigues et al., 2022; Sucunza et al., 2022). The alignment of our UAV transect monitoring results with patterns previously observed using traditional PAM monitoring within the same local area (Stedt et al., 2023) provides support and validation of the methods' robustness. For Bottlenose dolphins (*Tursiops truncatus*), Fettermann et al. (2022) found UAVs to increase detection probability and accuracy compared to a boat-based survey, with 26% more individuals identified and improved detection of individuals swimming directly below the water surface. Here we focused on harbour porpoises, but the UAV transect monitoring methodology is highly flexible and can be modified to allow monitoring of other surfacing aquatic species with different sizes, behaviours and ecology than our study species.

Limitations due to animal behaviour

UAV monitoring data allow for high-certainty group size estimation and behavioural and morphometric assessments (de Oliveira et al., 2023; Hodgson et al., 2017). To obtain absolute density estimates, the probability of detecting animals and how it may vary depending on behavioural changes needs to be estimated (Hammond et al., 2021; Macaulay et al., 2023). Behaviour may affect the probability of detecting animals in a spatio-temporally dependent manner, thus biasing density estimates across space and time. Porpoise dive behaviour depends on behavioural state (Macaulay et al., 2023) and therefore, surfacing rates may also bias the probability of detecting animals in non-acoustic surveys such as ours. As introduced earlier, perception bias, where animals are available for detection but missed, can be another major source of bias (Hammond et al., 2021). To obtain accurate and precise absolute density estimates, these biases need to be accounted for (Hammond et al., 2021).

We assumed no direct effect of the UAV on animal behaviour, especially since the operational altitude was higher than for most other studies investigating behavioural effects, but depending on context and species, this needs to be reconsidered (Bourke et al., 2023; Christiansen et al., 2016, 2020).

Limiting sampling conditions

Consistent with our finding of decreasing density from mornings to afternoons, previous studies investigating the diel rhythm of porpoises have reported high acoustic activity both at night, dusk and dawn (Carlström, 2005; Rojano-Doñate et al., 2024; Schaffeld et al., 2016; Stedt et al., 2023; Williamson et al., 2017). Nocturnal and crepuscular activity cannot be investigated using visual methods requiring daylight availability for data collection. Infrared cameras that detect warm air expelled as cetaceans breathe at the surface could however present a

© 2025 The Author(s). *Remote Sensing in Ecology and Conservation* published by John Wiley & Sons Ltd on behalf of Zoological Society of London.

20563485, 0, Downloaded from https://zslpublications.onlinelibrary.wiley.com/doi/10.1002/rse2.70043 by NICE, National Institute for Health and Care Excellence, Wiley Online Library on [25/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License



Figure 6. (a–d) Different weather conditions illustrating how porpoises appear in the footage. (a) A group of an estimated six porpoises, possibly in a feeding frenzy under an overcast sky. (b) Single porpoise without direct sunlight, appearing dark. (c) Single porpoise under clear sky, sunlight reflecting off the porpoise. (d) Single harbour *Phoca vitulina* or grey seal *Halichoerus grypus*, showing distinct morphological differences to porpoises.

possible solution, although these technologies are less accessible than commercial UAVs (Horton et al., 2019).

Both sea state and motor boats may have significantly reduced porpoise detections through availability and perception bias, especially at sea states of 3 and higher. It is unclear whether this was due to visibility or porpoise behaviour changes, which have been reported previously (Fig. 6a–c, Koski et al., 2009; Oliveira-Rodrigues et al., 2022; Sucunza et al., 2022). Teilmann (2003) shows a significant effect of sea state on relative harbour porpoise density. However, no difference in this effect was found between sea states 2 and 3, which deviates from our findings of decreased relative density for sea state 3. There are many possible explanations for this, such as differences in how observers assess sea state, the boat-based platform and other environmental factors affecting visibility.

Previous UAV monitoring studies of dugongs (*Dugong dugon*) and harbour porpoises showed no significant effects of environmental conditions, such as sea state and lighting conditions, on detection probability (Aniceto et al., 2018; Hodgson et al., 2017). Corresponding to our findings, Koski et al. (2009) found significantly lower porpoise detections at Beaufort Sea states above 2. The short observation window of UAVs can cause a negative bias for group size in harbour porpoises (Hodgson et al., 2017), with group sizes being underestimated as not all individuals in a group might be detectable at the surface by the UAV during the time of data collection. However, the use of UAV videos provides an opportunity for true group size estimations as indefinite detailed review of raw data is possible and groups can be observed for longer periods of time allowing detection of individuals as they change positions within the group.

Counting error and false negatives

The mean swimming speeds of 1.3–1.5 m s^{-1} for porpoises (Otani et al., 2001) show that it is possible for porpoises to swim between sampling areas, as there was a

© 2025 The Author(s). *Remote Sensing in Ecology and Conservation* published by John Wiley & Sons Ltd on behalf of Zoological Society of London.

20563485, 0, Downloaded from https://zslpublications.onlinelibrary.wiley.com/doi/10.1002/rse2.70043 by NICE, National Institute for Health and Care Excellence, Wiley Online Library on [25/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

30–60-min time window between sampling flights. For porpoises and other species where identification of individual animals is difficult, the potential re-sampling of same individuals that move between sampling areas is difficult to account for. Identification of individual animals could aid in obtaining absolute density estimates but also requires a species-specific trade-off between flight altitude (and therefore sampling effort) and recognition capability. For species where individual identification is possible, the monitoring method presented here may also be used for spatially explicit mark-recapture methods and to quantify the potential counting error caused by same-individual re-sampling. This can be done by interrupting the UAV autopilot upon sightings and decreasing the UAV altitude to a height allowing identification of individuals. Counting error could be reduced by having multiple pilots fly UAVs in different adjacent areas simultaneously, which would significantly reduce the risk of re-sampling fast-moving animals. Otherwise, gaps between sampling areas could mitigate this, but this limits the ability to detect fine-scale differences in a small area. In line with signal detection theory, there may be a trade-off between counting error and false negatives (missed detections), where reducing counting error by, for example, spacing out transects will also increase the risk of missing animals.

Observer bias and learning effect

Observer bias and different degrees of perception bias between observers are known to be a problem in both aerial and ship-based marine fauna surveys (Fuentes et al., 2015; Ronconi & Burger, 2009; Sucunza et al., 2022). It is often recommended to employ more than one observer during visual sampling and our results support this recommendation (Smyser et al., 2016). Due to perception bias, both independent observers missed the same number of porpoise observations, despite a difference in experience (one observer conducted two full reviews of 188 flights and a second observer a single review of 48 flights). However, the ability to quantify perception bias by employing additional independent observers represents a strength of our method. As new detections between independent observers plateau after multiple reviews, a total number of available animals and the degree of perception bias could be estimated. The absence of any learning effect implies that an independent observer does not require elaborate training. Mixed-species feeding aggregations resulted in three instances of the two observers deviating by 1–2 porpoises in their group size count (Fig. 6a), so animal behaviour may be an important factor in the detectability of individuals and groups (Stedt et al., 2024).

Potential modifications and improvements

UAV transect design should be well informed by the study goals, study area scale and size as well as target species size and potential for confusion with other species, surfacing behaviour and visibility. Pre-programmed flight paths allow the UAV pilot to fully focus on observing the camera feed and detecting porpoises during a flight. Flying at a higher altitude and faster speeds enables a much larger sampling area to be covered, but detectability may be decreased as target species appear smaller on the footage and less time is spent above every part of the water's surface. One possible approach is to fly at an increased altitude while having a significant transect overlap to increase the detection probability and account for sun glare (Aniceto et al., 2018; Hodgson et al., 2017). In general, adding a covariate for glare intensity, coverage and direction to model the effect may be more informative than our less direct variable of cloud cover. Further work on the effect of overlap on detection probability and reducing effects of glare is also recommended. To obtain a more accurate count of adults and calves, one could also pause the programmed UAV flight path once a detection is made to follow the porpoises for a specific time and then resume the flight path. Alternatively, a second UAV could be used to investigate observations more closely, allowing the primary UAV to continue the monitoring effort on pre-programmed transects. This way the method can be modified to find porpoises to perform a longer focal follow to capture behaviour, group size and structure, morphometrics and body condition comparisons between areas over time (Brown et al., 2023; Elliser et al., 2022). By continued long-term monitoring, it may be possible to build an inventory of individuals and groups that are commonly found in an area, track individual movement on a small scale, obtain better absolute abundance estimates and quantify counting error by measuring the number of times the same individual is encountered, which also has the potential for application of mark-recapture frameworks for density estimation (Koski & Young, 2022).

In this study, we were able to distinguish between marine mammal species in the study area with high certainty as seals and harbour porpoises are morphologically distinct, despite being similar in size, even at low image resolution (Fig. 6d). For monitoring of other species and communities, the flight altitude might however need to be adapted.

Overall, the limitations and benefits of our proposed UAV transect monitoring method should be explored further by, for example, cross-calibration with traditional methods. This may involve flying manned aircraft or boat-based strip transects concurrently with the UAV at a

© 2025 The Author(s). *Remote Sensing in Ecology and Conservation* published by John Wiley & Sons Ltd on behalf of Zoological Society of London.

20563485, 0, Downloaded from https://zslpublications.onlinelibrary.wiley.com/doi/10.1002/rse2.70043 by NICE, National Institute for Health and Care Excellence, Wiley Online Library on [25/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

similar speed, as described in Hodgson et al. (2023) for manned aerial survey cross-calibration with UAV transects.

Considering the size and conservation status of our study area and species, a non-invasive method on the scale of a few km^2 is recommended instead. A theodolite-based method such as described in Hodgson et al. (2017) may therefore be suited to mitigate bias and disturbance via motorboat and aircraft noise. A second or multiple observers would scan the water surface surveyed by the UAV concurrently using binoculars and recording locations using a theodolite. By cross-referencing the locations of animals in the theodolite and UAV footage, the probability of detecting animals via UAV could be elucidated further relative to established methods with better-known detection probabilities according to the framework described in Hodgson et al. (2017).

Conclusions

The proposed UAV transect monitoring method provides many advantages compared to traditional aerial and boat surveys, especially at smaller spatial scales, and offers an important methodological contribution in studies of marine megafauna, including but not limited to elusive species in need of conservation actions, like the harbour porpoise (Owen et al., 2024). We were able to calculate relative density estimates and detect small-scale spatio-temporal differences in the density of harbour porpoises with comparatively low effort. The ability to assess perception bias and observer bias, as well as mitigate fatigue and learning effects validates this method despite its novelty. In addition, data collection and video review do not require experienced observers with biological expertise, which provides versatility to this method and greater ease in performing multiple reviews by different observers. Limitations do however include availability bias, dependency on weather conditions that allow sampling without increased perception bias, and restricted spatio-temporal coverage. However, depending on the goals of a survey the restricted spatio-temporal coverage can also represent a methodological strength, as small- and micro-scale surveys that require high precision may instead benefit from the detailed UAV transect monitoring data.

The UAV transect monitoring method provides many advantages as an accessible, versatile and adaptable tool for standardized monitoring of aquatic surfacing species compared to traditional aerial and boat surveys, especially at smaller spatial scales. In addition, it is easily modifiable to fit the needs of different species and research questions. In coastal areas, where marine wildlife and human activities often co-exist, the proposed method offers a valuable contribution to monitoring and research aimed at investigating and reducing associated conflicts.

Acknowledgements

A special thank you to Per Carlsson for continuous guidance and feedback during this study. A special thank you also to the staff at naturum Kullaberg and the County Administrative Board of Skåne for practical support during field work. Thanks to the staff at the statistics help desk at the Department of Biology, Lund University, especially Nikolay Oskolkov and Rachel Steward, for useful advice regarding statistical analysis. Thanks also to Anders Nilsson for providing valuable feedback on the presentation of statistical results. This work was supported by the Royal Physiographic Society of Lund (grant no. 39589 to JS) and The Royal Swedish Academy of Sciences (grant no. BS2018-0080 to JS). UAVs were operated by licensed pilots under permission issued by the County Administrative Board of Skåne to JS, allowing flights in the Kullaberg nature reserve for research purposes (permission no. 521153692021).

Author Contributions

Dinah Hartmann: Conceptualization; investigation; writing – original draft; methodology; visualization; writing – review and editing; formal analysis; data curation. **Valdemar Palmqvist:** Validation; writing – review and editing. **Johanna Stedt:** Conceptualization; funding acquisition; writing – original draft; visualization; writing – review and editing; project administration; supervision; resources.

Conflict of Interest

The authors have no conflicts of interest to declare.

Data Availability Statement

The dataset obtained by manual video review of unmanned aerial vehicle (UAV) data is available through Mendeley data: https://data.mendeley.com/datasets/7mvydssb2x/1.

References

2008/56/EC. Directive 2008/56—Framework for community action in the field of marine environmental policy (Marine Strategy Framework Directive)—EU monitor. Retrieved from https://www.eumonitor.eu/9353000/1/j4nvk6yhcbpeywk_j9vvik7m1c3gyxp/vitgbgiogxml

92/43/EEG.n.d. Council Directive 92/43/EEC of 21 May 1992 on the conservation of natural habitats and of wild fauna and flora. Retrieved from https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:31992L0043

© 2025 The Author(s). *Remote Sensing in Ecology and Conservation* published by John Wiley & Sons Ltd on behalf of Zoological Society of London.

20563485, 0, Downloaded from https://zslpublications.onlinelibrary.wiley.com/doi/10.1002/rse2.70043 by NICE, National Institute for Health and Care Excellence, Wiley Online Library on [25/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Amundin, M., Carlström, J., Thomas, L., Carlén, I., Koblitz, J., Teilmann, J. et al. (2022) Estimating the abundance of the critically endangered Baltic proper harbour porpoise (*Phocoena phocoena*) population using passive acoustic monitoring. *Ecology and Evolution*, **12**(2), e8554. https://doi.org/10.1002/ece3.8554

Anderson, K., Gonzalez, F. & Gaston, K.J. (2025) Drones in ecology: ten years back and forth. *Bioscience*, **75**(8), 664–680. https://doi.org/10.1093/biosci/biaf069

Angliss, R.P., Ferguson, M.C., Hall, P., Helker, V., Kennedy, A. & Sformo, T. (2018) Comparing manned to unmanned aerial surveys for cetacean monitoring in the Arctic: methods and operational results. *Journal of Unmanned Vehicle Systems*, **6**(3), 109–127. https://doi.org/10.1139/juvs-2018-0001

Aniceto, A.S., Biuw, M., Lindstrøm, U., Solbø, S.A., Broms, F. & Carroll, J. (2018) Monitoring marine mammals using unmanned aerial vehicles: quantifying detection certainty. *Ecosphere*, **9**(3), e02122. https://doi.org/10.1002/ecs2.2122

Ayres, K.A., Ketchum, J.T., González-Armas, R., Galván-Magaña, F., Hearn, A., Elorriaga-Verplancken, F.R. et al. (2021) The use of an unoccupied aerial vehicle to survey shark species over sand and rocky-reef habitats in a marine protected area. *Journal of Fish Biology*, **99**(5), 1735–1740. https://doi.org/10.1111/jfb.14838

Bates, D., Mächler, M., Bolker, B. & Walker, S. (2015) Fitting linear mixed-effects models using lme4. *Journal of Statistical Software*, **67**(1), 1–48. https://doi.org/10.18637/jss.v067.i01

Benjamins, S., van Geel, N., Hastie, G., Elliott, J. & Wilson, B. (2017) Harbour porpoise distribution can vary at small spatiotemporal scales in energetic habitats. *Deep Sea Research Part II: Topical Studies in Oceanography*, **141**, 191–202. https://doi.org/10.1016/j.dsr2.2016.07.002

Bourke, E., Raoult, V., Williamson, J.E. & Gaston, T.F. (2023) Estuary stingray (*Dasyatis fluviorum*) behaviour does not change in response to drone altitude. *DRONES*, **7**(3), 164. https://doi.org/10.3390/drones7030164

Brandt, M.J., Höschle, C., Diederichs, A., Betke, K., Matuschek, R. & Nehls, G. (2013) Seal scarers as a tool to deter harbour porpoises from offshore construction sites. *Marine Ecology Progress Series*, **475**, 291–302. https://doi.org/10.3354/meps10100

Brennecke, D., Siebert, U., Kindt-Larsen, L., Midtiby, H.S., Egemose, H.D., Ortiz, S.T. et al. (2022) The fine-scale behavior of harbor porpoises towards pingers. *Fisheries Research*, **255**, 106437. https://doi.org/10.1016/j.fishres.2022.106437

Brown, A.M., Allen, S.J., Kelly, N. & Hodgson, A.J. (2023) Using unoccupied aerial vehicles to estimate availability and group size error for aerial surveys of coastal dolphins. *Remote Sensing in Ecology and Conservation*, **9**(3), 340–353. https://doi.org/10.1002/rse2.313

Brownell, R.L., Jr., Reeves, R.R., Read, A.J., Smith, B.D., Thomas, P.O., Ralls, K. et al. (2019) Bycatch in gillnet fisheries threatens critically endangered small cetaceans and other aquatic megafauna. *Endangered Species Research*, **40**, 285–296. https://doi.org/10.3354/esr00994

Buckland, S.T., Anderson, D.R., Burnham, K.P., Laake, J.L., Borchers, D.L., Thomas, L. et al. (2001) *Introduction to distance sampling: estimating abundance of biological populations*. Oxford: Oxford University Press.

Burnham, K.P. & Anderson, D.R. (2002) *Model selection and multimodel inference*. New York, NY: Springer. https://doi.org/10.1007/b97636

Butcher, P., Colefax, A., Gorkin, R., Kajiura, S., López, N., Mourier, J. et al. (2021) The drone revolution of shark science: a review. *Drones*, **5**(1), 8. https://doi.org/10.3390/drones5010008

Camphuysen, C. & Krop, A. (2011) Maternal care, calf-training and site fidelity in a wild harbour porpoise in the North Sea. *Lutra*, **54**(2), 123–126.

Carlström, J. (2005) Diel variation in echolocation behavior of Wild Harbor porpoises. *Marine Mammal Science*, **21**(1), 1–12. https://doi.org/10.1111/j.1748-7692.2005.tb01204.x

Christiansen, F., Nielsen, M.L.K., Charlton, C., Bejder, L. & Madsen, P.T. (2020) Southern right whales show no behavioral response to low noise levels from a nearby unmanned aerial vehicle. *Marine Mammal Science*, **36**(3), 953–963. https://doi.org/10.1111/mms.12699

Christiansen, F., Rojano-Doñate, L., Madsen, P.T. & Bejder, L. (2016) Noise levels of multi-rotor unmanned aerial vehicles with implications for potential underwater impacts on marine mammals. *Frontiers in Marine Science*, **3**. https://www.frontiersin.org/articles/10.3389/fmars.2016.00277

Clausen, K., Wahlberg, M., Beedholm, K., Deruiter, S. & Madsen, P. (2011) Click communication in harbor porpoises *Phocoena phocoena*. *Bioacoustics-the International Journal of Animal Sound and Its Recording - BIOACOUSTICS*, **20**, 1–28. https://doi.org/10.1080/09524622.2011.9753630

CMS. 2024 Governments adopt key measures for the conservation of dolphins, porpoises and other ocean species [Press release] https://www.cms.int/en/news/governments-adopt-key-measures-conservation-dolphins-porpoises-and-other-ocean-species

Colefax, A.P., Butcher, P.A. & Kelaher, B.P. (2018) The potential for unmanned aerial vehicles (UAVs) to conduct marine fauna surveys in place of manned aircraft. *ICES Journal of Marine Science*, **75**(1), 1–8. https://doi.org/10.1093/icesjms/fsx100

de Oliveira, L.L., Andriolo, A., Cremer, M.J. & Zerbini, A.N. (2023) Aerial photogrammetry techniques using drones to estimate morphometric measurements and body condition in south American small cetaceans. *Marine Mammal Science*, **39**(3), 811–829. https://doi.org/10.1111/mms.13011

Dinkel, T.M., Girard, A., Bär, T., Dähne, M., Craul, A.-K., Cosentino, M. et al. (2025) Performance of theodolites versus drones in land-based studies of marine mammals.

 © 2025 The Author(s). *Remote Sensing in Ecology and Conservation* published by John Wiley & Sons Ltd on behalf of Zoological Society of London.

20563485, 0, Downloaded from https://zslpublications.onlinelibrary.wiley.com/doi/10.1002/rse2.70043 by NICE, National Institute for Health and Care Excellence, Wiley Online Library on [25/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Scientific Reports, **15**(1), 20302. https://doi.org/10.1038/s41598-025-06978-8

Dyndo, M., Wisniewska, D.M., Rojano-Donate, L. & Madsen, P.T. (2015) Harbour porpoises react to low levels of high frequency vessel noise. *Scientific Reports*, **5**, 11083. https://doi.org/10.1038/srep11083

Elliser, C.R., van der Linde, K. & MacIver, K. (2022) Adapting photo-identification methods to study poorly marked cetaceans: a case study for common dolphins and harbor porpoises. *Mammalian Biology*, **102**(3), 811–827. https://doi.org/10.1007/s42991-021-00194-3

Fettermann, T., Fiori, L., Gillman, L., Stockin, K.A. & Bollard, B. (2022) Drone surveys are more accurate than boat-based surveys of bottlenose dolphins (*Tursiops truncates*). *Drones*, **6**(4), 82. https://doi.org/10.3390/drones6040082

Frankish, C.K., von Benda-Beckmann, A.M., Teilmann, J., Tougaard, J., Dietz, R., Sveegaard, S. et al. (2023) Ship noise causes tagged harbour porpoises to change direction or dive deeper. *Marine Pollution Bulletin*, **197**, 115755. https://doi.org/10.1016/j.marpolbul.2023.115755

Frouin-Mouy, H., Tenorio-Hallé, L., Thode, A., Swartz, S. & Urbán, J. (2020) Using two drones to simultaneously monitor visual and acoustic behaviour of gray whales (*Eschrichtius robustus*) in Baja California, Mexico. *Journal of Experimental Marine Biology and Ecology*, **525**, 151321. https://doi.org/10.1016/j.jembe.2020.151321

Fuentes, M.M.P.B., Bell, I., Hagihara, R., Hamann, M., Hazel, J., Huth, A. et al. (2015) Improving in-water estimates of marine turtle abundance by adjusting aerial survey counts for perception and availability biases. *Journal of Experimental Marine Biology and Ecology*, **471**, 77–83. https://doi.org/10.1016/j.jembe.2015.05.003

Gallagher, C.A., Chimienti, M., Grimm, V. & Nabe-Nielsen, J. (2021) Energy-mediated responses to changing prey size and distribution in marine top predator movements and population dynamics. *Journal of Animal Ecology*, **91**, 241–254. https://doi.org/10.1111/1365-2656.13627

Gilles, A., Authier, M., Ramirez-Martinez, N.C., Araújo, H., Blanchard, A., Carlström, J. et al. (2023) Estimates of cetacean abundance in European Atlantic waters in summer 2022 from the SCANS-IV aerial and shipboard surveys. Final report published 29 September 2023. 64 pp. https://tinyurl.com/3ynt6swa

Gilles, A., Scheidat, M. & Siebert, U. (2009) Seasonal distribution of harbour porpoises and possible interference of offshore wind farms in the German North Sea. *Marine Ecology Progress Series*, **383**, 295–307. https://doi.org/10.3354/meps08020

Gilles, A., Viquerat, S., Becker, E.A., Forney, K.A., Geelhoed, S.C.V., Haelters, J. et al. (2016) Seasonal habitat-based density models for a marine top predator, the harbor porpoise, in a dynamic environment. *Ecosphere*, **7**(6), e01367. https://doi.org/10.1002/ecs2.1367

Hamel, H., Torres Ortiz, S. & Wahlberg, M. (2025) Long-term observations reveal short-term mother–calf affiliation in wild harbour porpoises. *Animal Behaviour*, **219**, 122992. https://doi.org/10.1016/j.anbehav.2024.10.006

Hammond, P.s., Berggren, P., Benke, H., Borchers, D.l., Collet, A., Heide-Jørgensen, M.p. et al. (2002) Abundance of harbour porpoise and other cetaceans in the North Sea and adjacent waters. *Journal of Applied Ecology*, **39**(2), 361–376. https://doi.org/10.1046/j.1365-2664.2002.00713.x

Hammond, P.S., Francis, T.B., Heinemann, D., Long, K.J., Moore, J.E., Punt, A.E. et al. (2021) Estimating the abundance of marine mammal populations. *Frontiers in Marine Science*, **8**. https://doi.org/10.3389/fmars.2021.735770

Hao, X., Hamel, H., Grandjean, C.H., Fedutin, I., Wahlberg, M., Frankish, C.K. et al. (2024) Harbour porpoises respond to recreational boats by speeding up and moving away from the boat path. *Ecology and Evolution*, **14**(5), e11433. https://doi.org/10.1002/ece3.11433

Hao, X. & Nabe-Nielsen, J. (2023) Distribution and speed of recreational boats in Danish waters based on coastal observations and satellite images: predicting where boats may affect harbour porpoises. *OCEAN & COASTAL MANAGEMENT*, **242**, 106721. https://doi.org/10.1016/j.ocecoaman.2023.106721

Hartmann D. 2025 UAV monitoring of Harbour Porpoises at Kullaberg Nature Reserve, Sweden [dataset] *Mendeley Data*, V1. https://doi.org/10.17632/7mvydssb2x.1

Hermannsen, L., Ladegaard, M., Tønnesen, P., Malinka, C., Beedholm, K., Tougaard, J. et al. (2025) High-frequency vessel noise can mask porpoise echolocation. *Journal of Experimental Biology*, **228**(6), jeb249963. https://doi.org/10.1242/jeb.249963

Heße, E., Boyi, J.O., Das, K., Jung, K., Lehnert, K., Piette, M. et al. (2025) A multi-method approach reveals long- and short-term dietary differences in individual harbour porpoises *Phocoena phocoena* in the southern North Sea. *Marine Ecology Progress Series*, **755**, 115–132. https://doi.org/10.3354/meps14787

Hodgson, A., Kelly, N. & Peel, D. (2013) Unmanned aerial vehicles (UAVs) for surveying marine Fauna: a dugong case study. *PLoS One*, **8**(11), e79556. https://doi.org/10.1371/journal.pone.0079556

Hodgson, A., Peel, D. & Kelly, N. (2017) Unmanned aerial vehicles for surveying marine fauna: assessing detection probability. *Ecological Applications*, **27**(4), 1253–1267.

Hodgson, A.J., Kelly, N. & Peel, D. (2023) Drone images afford more detections of marine wildlife than real-time observers during simultaneous large-scale surveys. *PeerJ*, **11**, e16186. https://doi.org/10.7717/peerj.16186

Horton, T.W., Hauser, N., Cassel, S., Klaus, K.F., Fettermann, T. & Key, N. (2019) Doctor drone: non-invasive measurement of humpback whale vital signs using unoccupied aerial system infrared thermography. *Frontiers*

© 2025 The Author(s). *Remote Sensing in Ecology and Conservation* published by John Wiley & Sons Ltd on behalf of Zoological Society of London.

20563485, 0, Downloaded from https://zslpublications.onlinelibrary.wiley.com/doi/10.1002/rse2.70043 by NICE, National Institute for Health and Care Excellence, Wiley Online Library on [25/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

in Marine Science, **6**. https://doi.org/10.3389/fmars.2019.00466

Jefferson, T.A., Smultea, M.A., Courbis, S.S. & Campbell, G.S. (2016) Harbor porpoise (*Phocoena phocoena*) recovery in the inland waters of Washington: estimates of density and abundance from aerial surveys, 2013–2015. *Canadian Journal of Zoology*, **94**(7), 505–515. https://doi.org/10.1139/cjz-2015-0236

Johnston, D.W., Dale, J., Murray, K., Josephson, E., Newton, E. & Wood, S. (2017) Comparing occupied and unoccupied aircraft surveys of wildlife populations: Assessing the gray seal (*Halichoerus grypus*) breeding colony on Muskeget Island, USA. *Journal of Unmanned Vehicle Systems*. https://doi.org/10.1139/juvs-2017-0012

Jones, A.R., Hosegood, P., Wynn, R.B., De Boer, M.N., Butler-Cowdry, S. & Embling, C.B. (2014) Fine-scale hydrodynamics influence the spatio-temporal distribution of harbour porpoises at a coastal hotspot. *Progress in Oceanography*, **128**, 30–48. https://doi.org/10.1016/j.pocean.2014.08.002

Koski, W., Allen, T., Ireland, D., Buck, G., Smith, P., Macrander, A. et al. (2009) Evaluation of an unmanned airborne system for monitoring marine mammals. *Aquatic Mammals*, **35**, 347–357347. https://doi.org/10.1578/AM.35.3.2009.347

Koski, W., Gamage, G., Davis, A.R., Mathews, T., LeBlanc, B. & Ferguson, S.H. (2015) Evaluation of UAS for photographic re-identification of bowhead whales, Balaena mysticetus. *Journal of Unmanned Vehicle Systems*, **3**(1), 22–29. https://doi.org/10.1139/juvs-2014-0014

Koski, W.R. & Young, B.G. (2022) A new scoring system for use in capture–recapture studies for bowhead whales photographed with drones. *Drone Systems and Applications*, **10**(1), 15–25. https://doi.org/10.1139/juvs-2021-0027

Lockyer, C. & Kinze, C. (2003) Status, ecology and life history of harbour porpoise (*Phocoena phocoena*), in Danish waters. *NAMMCO Scientific Publications*, **5**, 143. https://doi.org/10.7557/3.2745

Macaulay, J.D.J., Rojano-Doñate, L., Ladegaard, M., Tougaard, J., Teilmann, J., Marques, T.A. et al. (2023) Implications of porpoise echolocation and dive behaviour on passive acoustic monitoring. *The Journal of the Acoustical Society of America*, **154**(4), 1982–1995. https://doi.org/10.1121/10.0021163

Marques, T.A., Thomas, L., Martin, S.W., Mellinger, D.K., Ward, J.A., Moretti, D.J. et al. (2013) Estimating animal population density using passive acoustics. *Biological Reviews*, **88**(2), 287–309. https://doi.org/10.1111/brv.12001

Nielsen, T., Wahlberg, M., Heikkilä, S., Jensen, M., Sabinsky, P. & Dabelsteen, T. (2012) Swimming patterns of wild harbour porpoises *Phocoena phocoena* show detection and avoidance of gillnets at very long ranges. *Marine Ecology Progress Series*, **453**, 241–248. https://doi.org/10.3354/meps09630

Norman, S., Hanson, M., Huggins, J., Lambourn, D., Calambokidis, J., Cottrell, P. et al. (2018) Conception, fetal growth, and calving seasonality of harbor porpoise (*Phocoena phocoena*) in the Salish Sea waters of Washington state, USA and southern British Columbia, Canada. *Canadian Journal of Zoology*, **96**, 566–575. https://doi.org/10.1139/cjz-2017-0155

Oliveira-Rodrigues, C., Correia, A.M., Valente, R., Gil, Á., Gandra, M., Liberal, M. et al. (2022) Assessing data bias in visual surveys from a cetacean monitoring programme. *Scientific Data*, **9**, 682. https://doi.org/10.1038/s41597-022-01803-7

Otani, S., Naito, Y., Kato, A. & Kawamura, A. (2001) Oxygen consumption and swim speed of the harbor porpoise *Phocoena phocoena*. *Fisheries Science*, **67**(5), 894–898. https://doi.org/10.1046/j.1444-2906.2001.00338.x

Owen, K., Gilles, A., Authier, M., Carlström, J., Genu, M., Kyhn, L.A. et al. (2024) A negative trend in abundance and an exceeded mortality limit call for conservation action for the Vulnerable Belt sea harbour porpoise population. *Frontiers in Marine Science*, **11**. https://doi.org/10.3389/fmars.2024.1289808

Pedersen, F. (1993) Fronts in the Kattegat—the hydrodynamic regulating factor for biology. *Estuaries*, **16**(1), 104–112. https://doi.org/10.2307/1352768

Ransom, J.I. (2012) Detection probability in aerial surveys of feral horses. *Journal of Wildlife Management*, **76**(2), 299–307. https://doi.org/10.1002/jwmg.204

Rees, A.F., Avens, L., Ballorain, K., Bevan, E., Broderick, A.C., Carthy, R.R. et al. (2018) The potential of unmanned aerial systems for sea turtle research and conservation: a review and future directions. *Endangered Species Research*, **35**, 81–100. https://doi.org/10.3354/esr00877

Rojano-Doñate, L., McDonald, B.I., Wisniewska, D.M., Johnson, M., Teilmann, J., Wahlberg, M. et al. (2018) High field metabolic rates of wild harbour porpoises. *The Journal of Experimental Biology*, **221**(23). https://doi.org/10.1242/jeb.185827

Rojano-Doñate, L., Teilmann, J., Wisniewska, D.M., Jensen, F.H., Siebert, U., McDonald, B.I. et al. (2024) Low hunting costs in an expensive marine mammal predator. *Science Advances*, **10**(20), eadj7132. https://doi.org/10.1126/sciadv.adj7132

Ronconi, R.A. & Burger, A.E. (2009) Estimating seabird densities from vessel transects: distance sampling and implications for strip transects. *Aquatic Biology*, **4**(3), 297–309. https://doi.org/10.3354/ab00112

Schaffeld, T., Bräger, S., Gallus, A., Dähne, M., Krügel, K., Herrmann, A. et al. (2016) Diel and seasonal patterns in acoustic presence and foraging behaviour of free-ranging harbour porpoises. *Marine Ecology Progress Series*, **547**, 257–272. https://doi.org/10.3354/meps11627

Scheidat, M., Verdaat, H. & Aarts, G. (2012) Using aerial surveys to estimate density and distribution of harbour

 © 2025 The Author(s). *Remote Sensing in Ecology and Conservation* published by John Wiley & Sons Ltd on behalf of Zoological Society of London.

20563485, 0, Downloaded from https://zslpublications.onlinelibrary.wiley.com/doi/10.1002/rse2.70043 by NICE, National Institute for Health and Care Excellence, Wiley Online Library on [25/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

porpoises in Dutch waters. *Journal of Sea Research*, **69**, 1–7. https://doi.org/10.1016/j.seares.2011.12.004

Smyser, T.J., Guenzel, R.J., Jacques, C.N. & Garton, E.O. (2016) Double-observer evaluation of pronghorn aerial line-transect surveys. *Wildlife Research*, **43**(6), 474–481. https://doi.org/10.1071/WR16006

Sonntag, R.P., Benke, H., Hiby, A.R., Lick, R. & Adelung, D. (1999) Identification of the first harbour porpoise (*Phocoena phocoena*) calving ground in the North Sea. *Journal of Sea Research*, **41**(3), 225–232. https://doi.org/10.1016/S1385-1101(98)00050-1

Stedt, J., Brokmar, L., Neimanis, A., Englund, W.E., Carlsson, P. & Roos, A. (2025) Combining DNA metabarcoding with macroscopic analysis increases the number of detected prey taxa in the estimated diet for harbour porpoises. *Frontiers in Marine Science*, **12**. https://doi.org/10.3389/fmars.2025.1517330

Stedt, J., Hamel, H., Torres Ortiz, S., Hojer Kristensen, J. & Wahlberg, M. (2024) Harbour porpoises are flexible predators displaying context-dependent foraging Behaviours. *Ecology and Evolution*, **14**(12), e70671. https://doi.org/10.1002/ece3.70671

Stedt, J., Wahlberg, M., Carlström, J., Nilsson, P.A., Amundin, M., Oskolkov, N. et al. (2023) Micro-scale spatial preference and temporal cyclicity linked to foraging in harbour porpoises. *Marine Ecology Progress Series*, **708**, 143–161. https://doi.org/10.3354/meps14268

Stepien, E.N., Nabe-Nielsen, J., Hansen, K.A., Kristensen, J.H., Blanchet, M.-A., Brando, S. et al. (2023) Determination of growth, mass, and body mass index of harbour porpoises (*Phocoena phocoena*): implications for conservational status assessment of populations. *Global Ecology and Conservation*, **42**, e02384. https://doi.org/10.1016/j.gecco.2023.e02384

Sucunza, F., Danilewicz, D., Andriolo, A., de Castro, F.R., Cremer, M., Denuncio, P. et al. (2022) Assessing bias in aerial surveys for cetaceans: results from experiments conducted with the franciscana dolphin. *Frontiers in Marine Science*, **9**. https://doi.org/10.3389/fmars.2022.1016444

Sveegaard S. 2011 Spatial and temporal distribution of harbour porpoises in relation to their prey. PhD thesis, Dep. of Arctic Environment, NERI. National Environmental Research Institute.

Sveegaard, S., Andreasen, H., Mouritsen, K.N., Jeppesen, J.P., Teilmann, J. & Kinze, C.C. (2012) Correlation between the seasonal distribution of harbour porpoises and their prey in the sound, Baltic Sea. *Marine Biology*, **159**(5), 1029–1037. https://doi.org/10.1007/s00227-012-1883-z

Sveegaard, S., Nabe-Nielsen, J., Stæhr, K.J., Jensen, T.F., Mouritsen, K.N. & Teilmann, J. (2012) Spatial interactions between marine predators and their prey: herring abundance as a driver for the distributions of mackerel and harbour porpoise. *Marine Ecology Progress Series*, **468**, 245–253. https://doi.org/10.3354/meps09959

Sweeney, K.L., Helker, V.T., Perryman, W.L., LeRoi, D.J., Fritz, L.W., Gelatt, T.S. et al. (2016) Flying beneath the clouds at the edge of the world: using a hexacopter to supplement abundance surveys of Steller sea lions (Eumetopias jubatus) in Alaska. *Journal of Unmanned Vehicle Systems*, **4**(1), 70–81. https://doi.org/10.1139/juvs-2015-0010

Teilmann, J. (2003) Influence of sea state on density estimates of harbour porpoises (Phocoena phocoena). *Journal of Cetacean Research and Management*, **5**(1), 85–92. https://doi.org/10.47536/jcrm.v5i1.830

Teilmann J. Dietz R. Sveegaard S. 2022 The use of marine waters in Skåne by harbour porpoises in time and space. Technical Report No. 236, DCE - Danish Centre for Environment and Energy.

Teilmann J. Sveegaard S. Dietz R. Petersen I. K. Berggren P. Desportes G. 2008 High density areas for harbour porpoises in Danish waters. NERI Technical Report No. 657, National Environemental Research Institute, University of Aarhus.

Thomsen, F., Laczny, M. & Piper, W. (2007) The harbour porpoise (*Phocoena phocoena*) in the central German bight: phenology, abundance and distribution in 2002-2004. *Helgoland Marine Research*, **61**(4), 283–289. https://doi.org/10.1007/s10152-007-0075-6

Till, J., Palmqvist, V., Wilk, E., Carlsson, P., & Stedt, J. (2025). Harbour porpoises in busy waters: Effect of recreational boats on swimming speed and surfacing interval investigated by two synchronised UAVs. https://doi.org/10.22541/au.176062531.12437747/v1

Unger, B., Nachtsheim, D., Martínez, N. R., Siebert, U., Kyhn, L. A., Balle, J. D., Teilmann, J., Carlström, J., Owen, K., & Gilles, A., 2021. MiniSCANS-II: Aerial survey for harbour porpoises in the western Baltic Sea, Belt Sea, the Sound and Kattegat in 2020. Danish Environmental Protection Agency, German Federal Agency for Nature Conservation and Swedish Agency for Marine and Water Management.

Verfuss, U.K., Aniceto, A.S., Harris, D.V., Gillespie, D., Fielding, S., Jiménez, G. et al. (2019) A review of unmanned vehicles for the detection and monitoring of marine fauna. *Marine Pollution Bulletin*, **140**, 17–29. https://doi.org/10.1016/j.marpolbul.2019.01.009

Viquerat, S., Herr, H., Gilles, A., Peschko, V., Siebert, U., Sveegaard, S. et al. (2014) Abundance of harbour porpoises (*Phocoena phocoena*) in the western Baltic, belt seas and Kattegat. *Marine Biology*, **161**(4), 745–754. https://doi.org/10.1007/s00227-013-2374-6

Webber, M., Keener, W., Wahlberg, M., Elliser, C., MacIver, K., Torres Ortiz, S. et al. (2023) Sexual Behavior and Anatomy in Porpoises. In: *Sex in cetaceans*. Cham: Springer, pp. 415–441. https://doi.org/10.1007/978-3-031-35651-3_18

Williamson, L.D., Brookes, K.L., Scott, B.E., Graham, I.M., Bradbury, G., Hammond, P.S. et al. (2016) Echolocation detections and digital video surveys provide reliable estimates of the relative density of harbour porpoises.

© 2025 The Author(s). *Remote Sensing in Ecology and Conservation* published by John Wiley & Sons Ltd on behalf of Zoological Society of London.

20563485, 0, Downloaded from https://zslpublications.onlinelibrary.wiley.com/doi/10.1002/rse2.70043 by NICE, National Institute for Health and Care Excellence, Wiley Online Library on [25/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Methods in Ecology and Evolution, **7**(7), 762–769. https://doi.org/10.1111/2041-210X.12538

Williamson, L.D., Brookes, K.L., Scott, B.E., Graham, I.M. & Thompson, P.M. (2017) Diurnal variation in harbour porpoise detection—potential implications for management. *Marine Ecology Progress Series*, **570**, 223–232. https://doi.org/10.3354/meps12118

Williamson, L.D., Scott, B.E., Laxton, M.R., Bachl, F.E., Illian, J.B., Brookes, K.L. et al. (2021) Spatiotemporal variation in harbor porpoise distribution and foraging across a landscape of fear. *Marine Mammal Science*, **38**(1), 42–57. https://doi.org/10.1111/mms.12839

Wisniewska, D.M., Johnson, M., Teilmann, J., Rojano-Donate, L., Shearer, J., Sveegaard, S. et al. (2016) Ultra-high foraging rates of harbor porpoises make them vulnerable to anthropogenic disturbance. *Current Biology*, **26**(11), 1441–1446. https://doi.org/10.1016/j.cub.2016.03.069

Wisniewska, D.M., Johnson, M., Teilmann, J., Siebert, U., Galatius, A., Dietz, R. et al. (2018) High rates of vessel noise disrupt foraging in wild harbour porpoises (*Phocoena phocoena*). *Proceedings of the Royal Society B: Biological Sciences*, **285**, 20172314. https://doi.org/10.1098/rspb.2017.2314

Supporting Information

Additional supporting information may be found online in the Supporting Information section at the end of the article.

Table S1 Scale of Beaufort Sea States 0–5 according to *Encylopædia Britannica*.

Table S2. Scale of Cloud Cover 1–4.

Table S3. Summary of the mean distance from the shore for each area with 95% confidence interval and the random effect intercept in the linear mixed model for each area.

Table S4. Summary statistics for individual predictors of the GLM for harbour porpoise counts by the second observer offset by sampling area size, including slope estimate, standard error, *z* value and Wald-statistic (*P*-value) for video order and the 'true' counts, which are assumed to be those of the first observer.

© 2025 The Author(s). *Remote Sensing in Ecology and Conservation* published by John Wiley & Sons Ltd on behalf of Zoological Society of London.

20563485, 0, Downloaded from https://zslpublications.onlinelibrary.wiley.com/doi/10.1002/rse2.70043 by NICE, National Institute for Health and Care Excellence, Wiley Online Library on [25/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License